Arista Access Node IND-ACC-65 Alarm Resolution SOP - Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding)

Node: IND-ACC-65 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_5540

Description

This failure domain occurs due to the inherent dynamic nature of routing protocols such as BGP, OSPF, and ISIS on the Arista Access Node IND-ACC-65. The protocols' continuous exchange of routing information and adjacency establishment lead to an unstable network topology, which in turn causes route churn. Additionally, the loss of OSPF adjacency and ISIS LSP flooding further exacerbate the situation. These protocol-related issues can arise due to various factors, including misconfiguration, hardware failures, or network congestion. As a result, the Arista Access Node IND-ACC-65 may generate the ALARM_5540, indicating a Warning severity. It is essential to identify and address the root cause of the issue to prevent network instability and potential service disruptions.

Pre-requisite Checks

- task: Verify the physical environment and ensure the Indiranagar Exchange is accessible
- task: Check the redundant link status to identify potential single points of failure
- task: Review the network topology and configuration to identify potential misconfigurations

Potential Root Causes

Cause	Probability
BGP route churn due to misconfigured AS-path filtering	High
OSPF adjacency loss due to incorrect network masking	Medium
ISIS LSP flooding due to hardware failure	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP summary information, Description: This command displays the BGP summary information, including the number of established peer connections, the number of prefixes received, and the number of prefixes advertised. It helps to identify potential issues with BGP route churn.

Command: show ospf neighbor, Expected Output: OSPF neighbor information, Description: This command displays the OSPF neighbor information, including the neighbor IP address, the neighbor state, and the number of LSA (Link-State Advertisement) received. It helps to identify potential issues with OSPF adjacency loss.
```

```
Command: show isis database, Expected Output: ISIS database information, Description: This command displays the ISIS database information, including the LSP (Link-State Packet) information, the SPF (Shortest Path First) tree, and the routing table. It helps to identify potential issues with ISIS LSP flooding.
```

Resolution Steps

Step 1: Step: Verify and correct BGP AS-path filtering, Command: show bgp as-path access-list, Description: Verify the AS-path access list configuration and correct any misconfigurations. This may involve adding or removing specific AS-numbers from the access list to prevent route churn.

Step 2: Step: Verify and correct OSPF network masking, Command: show ospf network, Description: Verify the OSPF network configuration and correct any misconfigurations. This may involve adjusting the network mask or adding or removing specific network addresses to prevent adjacency loss.

Step 3: Step: Verify and replace faulty hardware, Command: show system resource, Description: Verify the system resource utilization and identify any hardware issues. Replace any faulty hardware components to prevent ISIS LSP flooding.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP summary information, Description: Verify that the BGP route churn has been resolved and the BGP summary information is stable.

Command: show ospf neighbor, Expected Output: OSPF neighbor information, Description: Verify that the OSPF adjacency loss has been resolved and the OSPF neighbor information is stable.

Command: show isis database, Expected Output: ISIS database information, Description: Verify that the ISIS LSP flooding has been resolved and the ISIS database information is stable.
```

Roll-back Procedures

- Step: Restore the previous BGP AS-path filtering configuration, Command: config replace /etc/bgpd.conf , Description: Restore the previous BGP AS-path filtering configuration from a backup file to prevent route churn.

- Step: Restore the previous OSPF network masking configuration, Command: config replace /etc/ospfd.conf , Description: Restore the previous OSPF network masking configuration from a backup file to prevent adjacency loss.

- Step: Restore the previous ISIS LSP flooding mitigation configuration, Command: config replace /etc/isisd.conf , Description: Restore the previous ISIS LSP flooding mitigation configuration from a backup file to prevent ISIS LSP flooding.